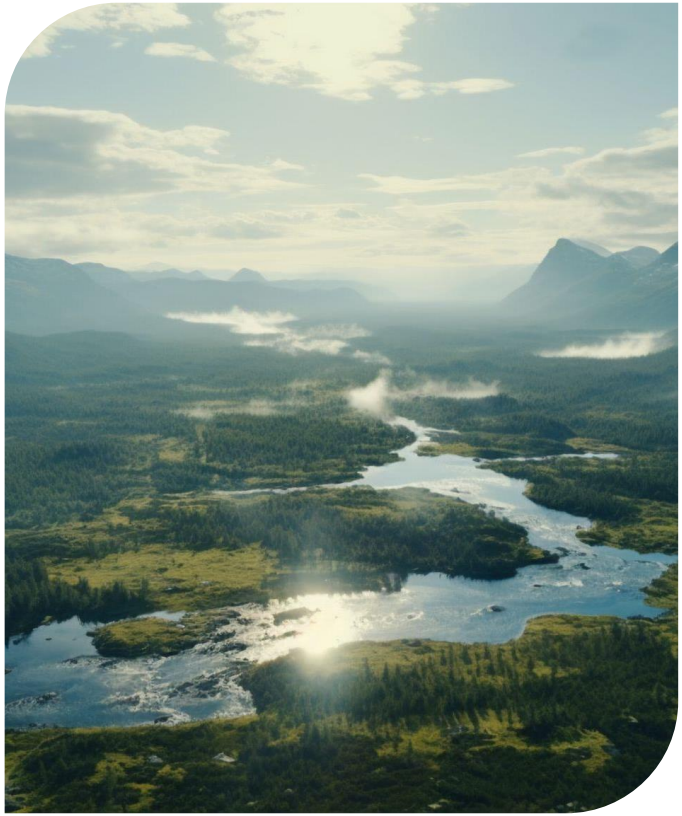
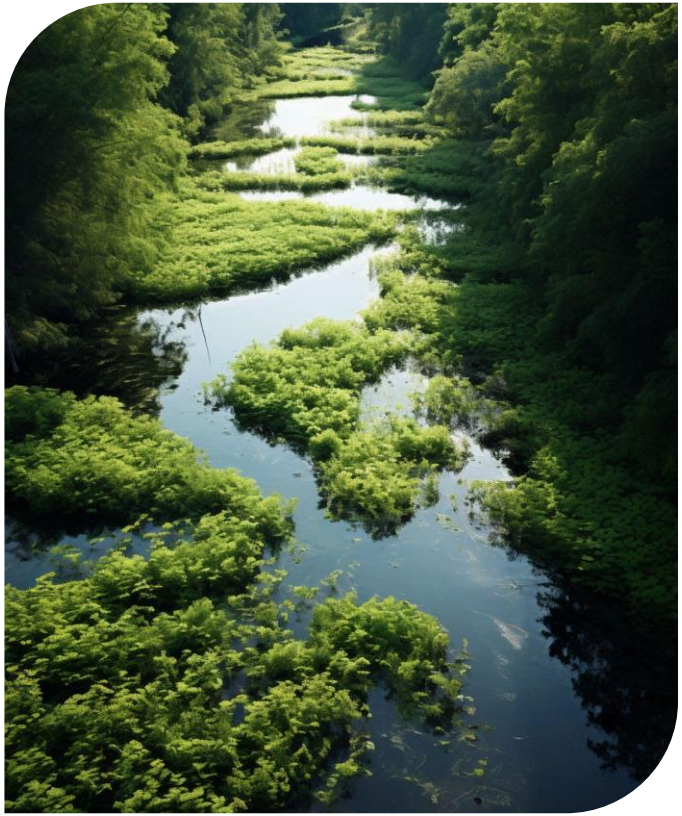
THE DESIGN AND IMPLEMENTATION OF ENVIRONMENTAL POLICY



Contents

- Criteria for Evaluating Environmental Policies
- Environmental Standards
- Pigouvian Taxes
- Effluent Fees
- Tradable Permits
- Quotas (Command and Control Instruments)
- Choice between Taxes and Quotas
- Environmental Policy: Regulation and Implementation



01

Criteria for Evaluating Environmental Policies

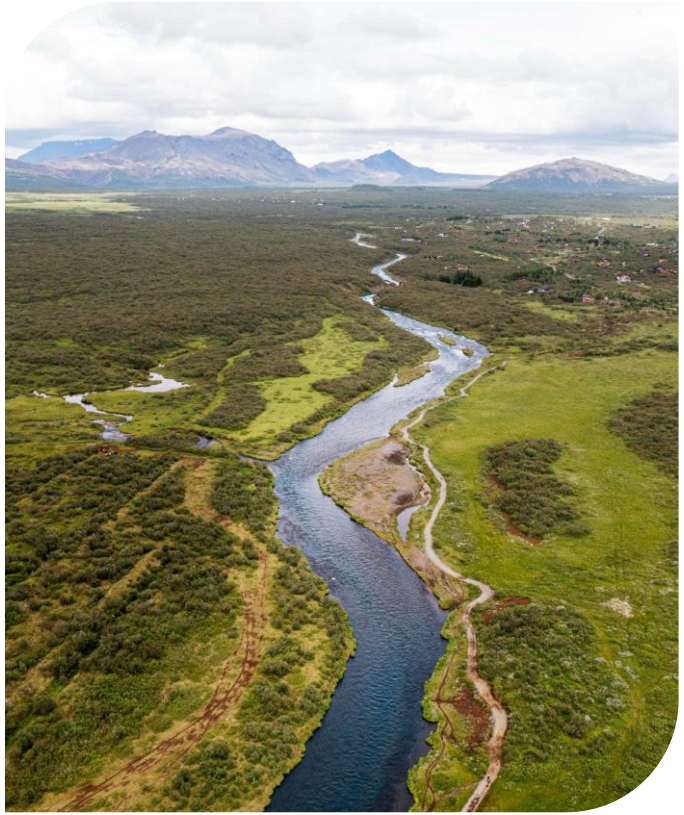
- Carbon Pricing
- Renewable Energy Incentives
- Energy Efficiency Standards
- Waste Management Regulations
- Protected Areas and Conservation
- Environmental Impact Assessments (EIAs)
- Water Management Policies
- Sustainable Agriculture Practices
- Transportation Policies
- Climate Adaptation Strategies
- Environmental Education and Awareness Programs
- International Agreements and Treaties:

1. Efficiency

- Administration in implementing various policies is said to be efficient when it gives maximum benefits to the society or when it adds into the well-being of the society compare to the social cost.

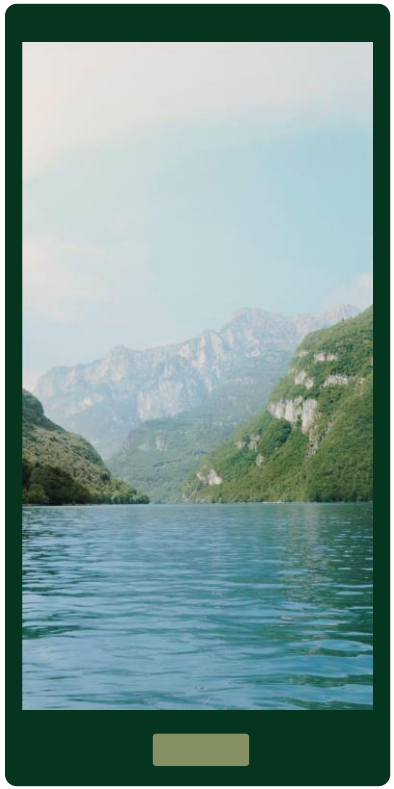
2. Cost Effective

Cost effectiveness criterion of environmental policy implies that the cost incurred to achieve environmental improvement should bring about a greater amount of environmental improvement at the least possible cost



3. Fairness

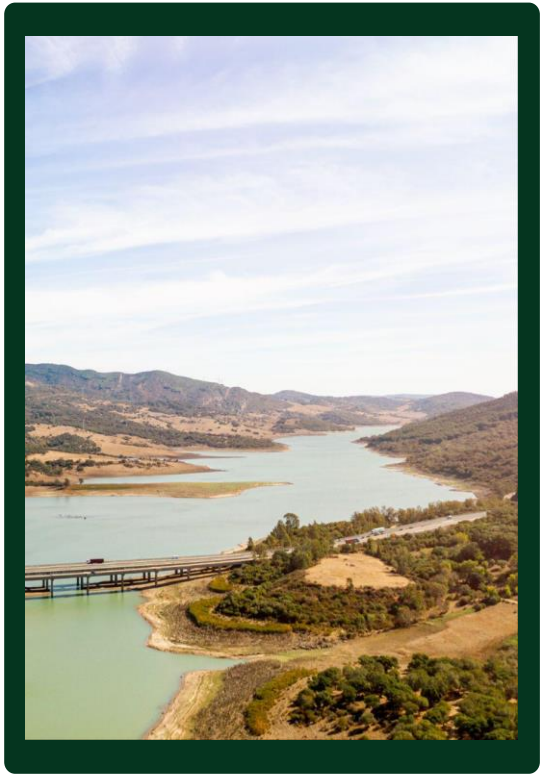
Fairness criterion for evaluating environmental policies is justifiable on the principle of equity. There should be equal distribution of benefits among the members of the society.



				Distribution of Net benefits	
Air pollution control program	Total cost	Total benefits	Net benefits	Region X	Region Y
A	50	100	50	25	25
B	50	100	50	30	20
C	50	140	90	20	70
D	50	140	90	40	50

The four alternative air pollution control programs that can be adopted by the government i.e. A, B, C, and D.

There are two regions X and Y where region X represents lower income group people and region Y represents high income group people.



4. Incentives for technological improvements

5. Enforceability

6. Moral consideration

Maybe you need to think about it!

Cement

In 2022, 350.6 Million tonnes (Infomerics Economic Research).
In 2027, cement consumption is anticipated to amount to 419.92 million tons.

With annual emissions of Sulphur dioxide, nitrogen oxide, and carbon monoxide over 500,000 tonnes, the cement industry ranks third among industrial polluters.

Semiconductor fabrication unit

According to Micro Commercial Components Corp. (MCC), fabricating semiconductor circuits on a 150-mm wafer uses 285 kWh of power or 1.6 kWh per square centimeter.
Moreover, it contributes to 31% of the global greenhouse gas emission. This is higher than what electricity generation contributes, which is 27%.

Maybe you need to think about it!

Fans

According to CEEW Survey about 40 million ceiling fans are sold in India every year.

The India ceiling fan market size reached 41.2 Million Units in 2022. Looking forward, IMARC Group expects the market to reach 47.5 Million Units by 2028,

E-waste

The volume of e-waste in India has witnessed a significant surge to **1.6 million tonnes in 2021-22.**

There is a need for strict adherence of environmental standards to check further deterioration of environment and simultaneously improving the environmental condition.

Environmental Standards

- Environmental standards refer to the maximum permissible concentration of harmful or potentially harmful pollutants which can be released that guarantees an acceptable health condition or environmental quality.

- Environmental standards help in setting a permissible limit of the pollutants to protect the human and environment health.

- In other words, these standards are numerical values and range of permissible limit prescribed by the governments across the globe and implemented by regulatory agencies in the respective countries to decide the environmental quality.

- This is generally expressed in terms of the permissible concentration of pollutants in unit of air emitted or in waste water discharge from source. It can also be expressed in terms of total load of pollutants per time unit, unit of production or unit of energy or materials inputs.

Types of Environmental Standards

Ambient

Emission

Technology

Ambient Standards

- An ambient environmental standards refers to the surrounding concentration of pollutants.
- Ambient standards expressed in terms of average concentration level or pollutants generated from various point and non-point sources.
- National Ambient Air Quality Standards set by Central Pollution Control Board is based on 24 hours and 8 hours.

Emission Standards

Emission standards are the permissible limits which never-exceeds beyond set limit applied directly coming from an industry or other sources.

It can be set on a wide variety of different bases.

For examples,

i) emission rate (e.g., kilograms per hour),

ii) emission concentration (e.g., parts per million of biochemical oxygen demand, or BOD, in waste-water),

iii) total quantity of residuals,

iv) residuals produced per unit of output (e.g., SO2 emissions per kilo watt hour of electricity produced, grams of CO per tonne of asphalt produced),

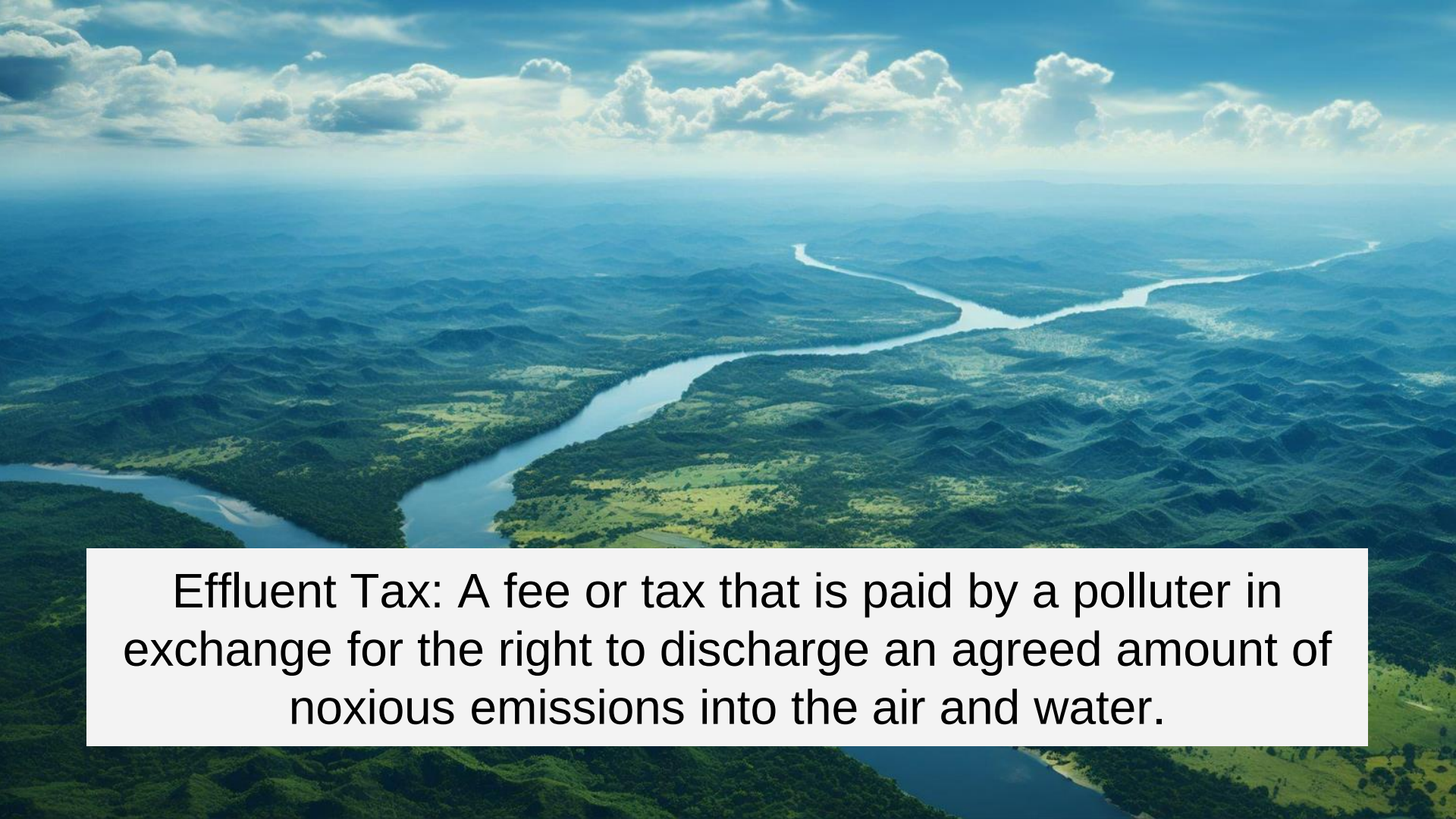
v) residual content per unit of input (e.g., sulphur content of coal used in power generation),

vi) Percentage removal of pollutants (e.g., 60-per-cent removal of waste materials before discharge).

Technology Standards

- Technology standards specify the technologies, techniques, or practices including design, engineering, input and output standards which polluters must adopt or meet to protect the environment during manufacturing stage.
- In contrast to emission standards, technology standards impose on polluters certain decisions and technologies to be used.
- This is some form of "technology forcing' for polluting industries to adopt technological change in order to meet environmental standards.

Pigouvian Tax



Effluent Tax: A fee or tax that is paid by a polluter in exchange for the right to discharge an agreed amount of noxious emissions into the air and water.

Effluent Fee

- Effluent fees should justify and reflect the environmental cost of pollution caused by effluents.
- Effluent fees should bear some relation to Marginal abatement cost borne by the producer so that the producer will be motivated to reduce Pollution
- Enough revenue should be generated for the maintenance and treatment of water bodies.
- Enough economic incentive in the form of offsetting of pollution charges to be given to the producer who releases clean effluent into the environment.

Tradable permits are used in an environmental regulatory scheme where the sources of the pollutant to be regulated are given permits to release a specified number of tons of the pollutant.

Tradable permits

- Tradable permits aim to reduce pollution emissions. There are two ways to issue these permits;

i) To auction it to the highest bidder; or

ii) To give it to the existing polluters in proportion to their baseline emissions.

- The permit owners then free to buy or sell to anybody i.e. to other polluters who wants to participate in this market.
- The basic purpose of the tradable permit is to raise revenue by selling the permits and to reduce pollution over a period of time.
- If a firm fails to abide by the permissible limit then it has to buy this from other firms or from the government. The price of this permit in the market is therefore determine by the demand for and supply of these permits.
- In the long run the government can reduce the size of tradable permits and by selling it at a higher rate can induce the firms to innovate or adopt the measures which reduces the pollution or emission and thereby to reduce the cost of this pollution

Problems of tradable pollution permits:

Difficult to determine the quantity of tradable permits.

Difficult to measure the level of the pollution by the firms

Administrating cost

Offsetting schemes

Political Interference

Global Pollution Permits

Quotas or Command and control instruments prescribes certain rules and standards to comply with by using sanctions.

It means setting up of a limit or permission for a certain amount of a targeted thing.

It is often used in Environmental legislation where policy makers intends to impose restrictions to ensure environmental sustainability.

https://youtu.be/UZqT509antY?si=UUNBTtZdEs0Uzgjt

Choice between taxes and quotas

Taxes are the direct payment in monetary terms made by the firms to government for emission of pollutants.

Measurement of taxes in proportion to the emission is difficult.

Direct positive impact on reduction of pollution

Whereas Quotas are the quantitative limits imposed on the emission of pollutants.

Quotas are easy to determine.

Does not have direct positive impact on reduction of pollution

ENVIRONMENTAL POLICY: REGULATION AND IMPLEMENTATION

1. Environment (Protection) Act, 1986
2. Water (Prevention and Control of Pollution) Act, 1974
3. Air (Prevention and Control of Pollution) Act, 1981
4. Forest Conservation Act, 1980
5. Wildlife (Protection) Act, 1972
6. National Environmental Tribunal Act, 1995
7. National Environment Appellate Authority Act, 1997
8. National Green Tribunal (NGT) Act, 2010
9. Paris Agreement (2015)
10. The Plastic Waste Management Rules, 2016
11. The Compensatory Afforestation Fund Act, 2016
12. The Wetlands (Conservation and Management) Rules, 2017
13. The Environmental Impact Assessment Notification, 2020

Summary

Topic Discussed

Criteria for Evaluating Environmental Policies; Environmental Standards; Pigouvian Taxes; Effluent Fees; Tradable Permits; Quotas (Command and Control Instruments); Choice between Taxes and Quotas; Environmental Policy: Regulation and Implementation

Assignment

Environmental Policy: Regulation and Implementation (MCQ TEST)

Activities

Effluent Fees; Tradable Permits

Thanks!

Do you have any questions?

Suggestions!

END OF EDUCATION IS CHARACTER!